

# Vasanthi Vaddelli

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## PROFESSIONAL SUMMARY

- Accomplished **Senior Software Engineer** with **8+ years of hands-on experience** designing, building, and deploying scalable, event-driven, and cloud-native systems across **healthcare, finance, and tech sectors**. Expert in **Python, Java, React.js**, and **cloud platforms** including **AWS and Azure**.
- Specialized in **microservices architecture, real-time data processing (Kafka, Spark), CI/CD automation (Jenkins, GitHub Actions, Azure DevOps)**, and **container orchestration** using **Docker** and **Kubernetes**.
- Proven success delivering high-impact solutions, such as **predictive ML models (XGBoost, TensorFlow)**, **LLM-powered development workflows (OpenAI API, LangChain, RAG)**, and cloud cost optimizations (**serverless, autoscaling**) that reduced operational overhead by up to **40 %**.
- Strong background in **full-stack development** and **performance tuning**, with significant achievements including **90 %+ test coverage**, latency reductions, and feature delivery ahead of schedule.
- Demonstrated leadership in **Agile teams**, driving **cross-functional collaboration, architectural modernization**, and productivity boosts through **automation, infrastructure-as-code**, and **modern DevOps best practices**.

## TECHNICAL SKILLS

- Programming Languages:** Java (8–21), Python (Pandas, NumPy), JavaScript (ES6+), TypeScript, SQL, C++, Bash
- Frontend Development:** React.js, Angular, Redux, HTML5, CSS3, Bootstrap, WebSocket, REST APIs
- Backend & Middleware:** Spring Boot, Spring MVC, Spring Security, Hibernate, JPA, Microservices Architecture, Node.js, Django, Flask, FastAPI, Apache Kafka, RabbitMQ, JMS
- Cloud Platforms & MLOps:** AWS (Lambda, EKS, S3, RDS, SageMaker), Microsoft Azure (Event Hubs, Azure ML, DevOps), Docker, Kubernetes, Terraform, Helm, Jenkins, GitHub Actions, CI/CD, Model Deployment & Monitoring
- Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras, BERT, NLP, Transformers, LLM Integration, LangChain, OpenAI API, RAG, SHAP, LIME, Hyperparameter Tuning, Feature Engineering, Prompt Engineering
- Big Data & Data Engineering:** Apache Spark, Hadoop, Apache Airflow, Data Modeling, PostgreSQL, MySQL, MongoDB, Oracle
- Architecture & Services:** RESTful APIs, GraphQL, gRPC, Service Discovery (Eureka), API Gateway, Circuit Breaker (Hystrix)
- DevOps & Monitoring:** Git, Bitbucket, Jenkins, Prometheus, Grafana, ELK Stack, Splunk
- Testing & Quality Assurance:** JUnit, TestNG, Mockito, Selenium, Cypress, Playwright, Pytest, unittest
- Messaging & Integration:** Apache Kafka, RabbitMQ, JMS, Protocol Buffers
- Methodologies & Tools:** Agile/Scrum, SAFe, Jira, Confluence, Cross-Functional Collaboration, Technical Leadership

## PROFESSIONAL EXPERIENCE

### CVS Health Corporation | AZ

June 2025 – Present

#### Senior Software Engineer

- Engineered event-driven **microservices** using **Python (FastAPI/Flask)**, **Apache Kafka**, and **Kubernetes** to process high-volume healthcare transactions, achieving **99.99 % uptime** and horizontal scalability for peak demand periods.
- Automated **CI/CD pipelines** with **Jenkins** and **GitHub Actions**, reducing manual deployment efforts by **40 %** and accelerating release cycles.
- Designed and deployed **AWS cloud infrastructure (Lambda, EKS, S3)** to optimize claims processing workflows, reducing operational costs by **25 %** through **serverless automation** and intelligent auto-scaling policies.
- Developed **predictive models** using **XGBoost** and **TensorFlow** to **forecast medication adherence risks**, achieving **92 % accuracy** in identifying high-risk patients for targeted interventions.

- Architected **LLM-powered code assistance and documentation workflows** using **OpenAI's API**, **LangChain**, and **RAG pipelines** to automate **documentation generation** and **code reviews**, enhancing developer productivity while maintaining **code quality standards**.
- Created comprehensive **unit**, **integration**, and **end-to-end test suites** using **pytest**, **unittest**, **Selenium**, and **Playwright (Python)**, increasing test coverage from **65 % to 90 %** and reducing production issues by **40 %**.
- Designed and optimized **PostgreSQL** and **MySQL database schemas**, applying **indexing** and normalization strategies to improve query performance by **35 %** and reduce storage costs by **20 %**.
- Led a cross-functional team of **5 engineers** in modernizing the pharmacy benefits management platform, coordinating architecture decisions and **sprint planning** to deliver features 2 weeks ahead of schedule.

## Microsoft | India

March 2021 – December 2022

### Software Engineer

- Designed and implemented **scalable cloud services** using **Java**, **Spring Boot**, and **Microsoft Azure**, increasing system throughput by **40 %** and supporting over **2 million monthly active users**.
- Optimized **real-time data pipelines** with **Azure Event Hubs** and **Apache Spark**, reducing data processing latency by **50 %** for **mission-critical analytics** workloads.
- Delivered **full-stack features** using **React.js** and **TypeScript** on the **frontend** and **Java/Spring Boot** on the **backend** for internal productivity tools, improving **team efficiency** by **25 %**.
- Optimized a **BERT-based NLP model** for **Azure Cognitive Services** using quantization and **ONNX runtime**, significantly improving inference speed while preserving high accuracy for text processing workloads.
- Integrated **LLM-powered features** using **OpenAI API** and **LangChain** into internal developer tooling, automating code review summaries and accelerating documentation workflows.
- Developed an automated **testing framework** with **JUnit**, **Mockito**, and **Selenium**, increasing test coverage from **65 % to 90 %** and reducing production bugs by **35 %**.
- Implemented automated **CI/CD pipelines** using **Azure DevOps**, decreasing manual **QA workload** by **60 %** and accelerating release timelines.
- Automated **infrastructure provisioning** using **Terraform** and **Azure Bicep**, reducing **deployment times** by **50 %** and supporting **multi-region scalability**.
- Led **Agile transformation** for **3 feature teams** using **Scrum** and **SAFe** practices, optimizing **Jira workflows**, **sprint retrospectives**, and planning processes to improve delivery speed by **30 %** and reduce sprint carryover by **20 %**.

## JPMorgan Chase | India

December 2018 – February 2021

### Software Engineer

- Developed core functionalities for a **Single Page Application (SPA)** using **JavaScript (ES6+)**, **React**, and **Redux**, improving **state management** and increasing **user engagement** by **12 %**.
- Designed and deployed **cloud-native microservices** using **Docker**, **Kubernetes**, **YAML manifests**, and **Helm charts** for **automated** and **scalable** deployments.
- Built and integrated **RESTful APIs** using **Java 8** and **Spring Boot**, implementing **JWT authentication**, optimized **payloads**, and **caching strategies** to improve data transfer efficiency by **18 %**.
- Implemented high-performance **inter-service communication** using **gRPC** and **protocol buffers**, replacing **REST endpoints** for latency-sensitive trading systems to enable real-time data streaming between microservices.
- Achieved **98 % code coverage** by implementing robust **unit** and **integration testing** strategies using **JUnit**, **Mockito**, **Selenium**, and **Cypress**.
- Worked with **quantitative analysts** to optimize derivatives pricing models using **Monte Carlo simulations** with **Dask** and **Ray**, improving **P&L accuracy** by **15 %**.
- Engineered **Python-based data pipelines** with **Pandas**, **NumPy**, and **Apache Airflow** to automate daily **P&L reporting** and **trade reconciliation**, reducing manual workload by **65 %** and improving **compliance accuracy**.

## Oracle | India

May 2015 – December 2018

### Python Developer

- Developed and optimized backend services and **RESTful APIs** using **Python**, **Django**, and **Flask**, improving system performance by **30 %** and reducing response times by **40 %**.

- Designed and implemented **data processing scripts** using **Python, Pandas**, and **NumPy** for enterprise applications integrating with **Oracle databases**, streamlining data access and improving response times by **25 %**.
- Built automated **ETL pipelines** using **Python** and **SQL** to streamline data ingestion and reporting workflows, reducing manual data-handling effort by **35 %**.
- Increased **unit test coverage** from **60 % to 85 %** by implementing **automated testing frameworks** using **Pytest** and **unittest**, leading to a **35 % reduction** in post-release defects.
- Deployed applications using **Docker** and **Kubernetes** in collaboration with **DevOps teams**, cutting deployment time by **50 %** and enhancing **scalability**.
- Resolved **high-priority production issues** by analyzing logs using **SQL Profiler** and the **ELK Stack (Elasticsearch, Logstash, Kibana)**, reducing system downtime by **20 %**.

## EDUCATION

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- **Master's in Business Analytics**, University of New Haven, Connecticut — 07/2023 to 05/2025